CropSage Crop Catalog Schema

Human-readable table view of catalog.schema.json. The JSON Schema remains the machine validation contract; this PDF is its implementation and review companion.

Required-field rule: every crop record must contain all 28 finalized fields. Nullable heat and water values stay present and explicitly marked when numeric evidence has not been established.

Schema Overview

Property	Value	Meaning
Schema draft	https://json-schema.org/draft/2020-12/schema	JSON Schema dialect used by the machine contract.
Schema ID	https://cropsage.local/schemas/crop-catalog-1.0.0.json	Stable identifier for this schema version.
Catalog version	1.0.0	The schema currently accepts catalog version 1.0.0 only.
Crop count	exactly 22	The current catalog release must contain the selected 22 crop profiles.
Unknown fields	rejected	additionalProperties is false for catalog and crop objects.

Finalized Crop Fields

Group	Field	Type	Stored requirement	Location comparison / runtime use
Identity	crop_id	string	Stable machine identifier; unique snake_case key.	Catalog lookup and audit key.
Identity	common_name	string	Human-readable crop profile name.	Displayed in rankings and explanations.
Identity	scientific_name	string	Botanical taxon for the profile.	Prevents common-name ambiguity.
Identity	production_use	string	Harvest endpoint such as grain, silage, hay, lint, or fresh market.	Keeps requirements tied to the intended product.
Identity	variety_scope	string	Included cultivar or market classes and explicit exclusions.	Flags cultivar assumptions before scoring.
Region and season	supported_texas_regions	region_id[]	Regions where the crop can enter recommendation ranking.	Compared with the resolved Texas region.
Region and season	regional_suitability	object[5]	Rating and basis for every Texas catalog region.	Provides regional prior and explanation.
Region and season	planting_windows_by_region	object[]	Month-level planting windows, evidence status, and sources.	Compared with target date, frost outlook, and soil temperature.
Region and season	days_to_maturity	object	Minimum and maximum calendar days plus basis and evidence.	Compared with frost-free window and forecast horizon.
Temperature	optimal_temperature_range	object	Minimum and maximum degrees C, measurement basis, evidence, and sources.	Compared with FortyGuard forecast and NASA climatology.
Temperature	heat_stress_threshold	object	Optional numeric warning threshold with operator and scoring permission.	Compared with stage-aware forecast/climatology heat evidence.
Temperature	heat_exposure_duration	object	Optional hours or consecutive days; null when not established.	Controls whether a heat event can receive a penalty.
Temperature	heat_sensitive_stages	string[]	Growth stages most vulnerable to heat.	Matched to estimated crop stage before heat scoring.
Temperature	frost_sensitivity	object	Frost class, explanation, and sources.	Compared with forecast minimum and frost-free season.
Temperature	temperature_unit	constant	Always degC in the catalog.	Prevents conversion ambiguity.

Group	Field	Type	Stored requirement	Location comparison / runtime use
Temperature	temperature_measurement_basis	string	Air, canopy, soil, daily maximum, daily mean, or stage context.	Selects the correct location variable for comparison.
Soil	preferred_soil_textures	string[]	Acceptable/preferred USDA-style texture names.	Compared with dominant SSURGO component texture.
Soil	ph_tolerable_range	object	Minimum and maximum soil pH with evidence metadata.	Compared with SSURGO pH; lab sample overrides mapped estimate.
Soil	drainage_requirement	object	Drainage class and agronomic explanation.	Compared with SSURGO drainage class and ponding/flooding properties.
Water	water_demand	object	Demand class and optional seasonal range in millimeters.	Compared approximately with precipitation plus soil storage and irrigation.
Water	drought_tolerance	enum	Relative tolerance from very_low to very_high.	Modifies water-deficit risk, never replaces water balance.
Water	irrigation_requirement	object	Operational class, explanation, and sources.	Compared with farmer-reported irrigation availability and reliability.
Evidence	source_ids	SRC-ID[]	Union of references supporting the record.	Resolves into the catalog reference registry.
Evidence	confidence	enum	High, medium, or low evidence confidence.	Displayed and used to cap recommendation confidence.
Evidence	last_reviewed	date	Date the record was last reviewed.	Drives review-age warnings.
Evidence	assumptions	string[]	Known simplifications and required local checks.	Included in explanations and audit output.
Administration	catalog_version	constant	Version 1.0.0 for every current record.	Pins a recommendation to exact catalog content.
Administration	record_status	enum	Provisional, approved, or retired.	Only approved records should be eligible for production hard scoring.

Nested Object Definitions

Object	Required members	Purpose
temperatureRange	min_c, max_c, basis, evidence_status, source_ids	Numeric range plus measurement interpretation and provenance.
heatThreshold	value_c, operator, basis, severity, scoring_use, evidence_status, source_ids	Nullable threshold. Informational-only values cannot produce a penalty.
heatDuration	hours, consecutive_days, description, scoring_use, evidence_status, source_ids	Nullable duration. Missing numeric evidence remains explicit.
phRange	min, max, evidence_status, source_ids	Soil pH bounds and provenance.
waterDemand	class, seasonal_range_mm, basis, evidence_status, source_ids	Qualitative class with optional millimeter range.
regionalSuitability	region_id, rating, basis	One rating for each of the five Texas regions.
plantingWindow	region_id, windows, basis, evidence_status, source_ids	One or more month ranges for a region.
daysToMaturity	min, max, basis, evidence_status, source_ids	Calendar-day range and interpretation.
frostSensitivity	class, description, source_ids	Relative frost response and explanation.
drainageRequirement	class, description, source_ids	Required drainage behavior and explanation.
irrigationRequirement	class, description, source_ids	Operational irrigation need and explanation.

Allowed Values

Vocabulary	Allowed values
region_id	plains; eastern; central_south_winter_garden; lower_rio_grande_valley; far_west
evidence_status	direct; regional_transfer; generalized; not_numerically_established
regional rating	primary; suitable; conditional; limited; not_supported
scoring_use	soft_penalty; informational_only
water demand	low; moderate; high; very_high
drought tolerance	very_low; low; moderate; high; very_high
irrigation class	usually_required; recommended; conditional; often_rainfed
drainage class	poorly_drained_tolerant; moderate; well_drained; very_well_drained
frost class	very_high; high; moderate; low; dormant_hardy
confidence	high; medium; low
record status	provisional; approved; retired

Evidence and Scoring Rules

Evidence status	Permitted deterministic behavior
direct	May contribute to deterministic scoring when the field also permits soft_penalty and the measurement basis matches runtime evidence.
regional_transfer	Soft evidence only. It cannot independently justify a statewide hard exclusion.
generalized	Informational or low-weight soft evidence only.
not_numerically_established	Never triggers a hard filter or numeric penalty. It must remain visible as an evidence gap.
provisional record	Suitable for development and demonstration. Production hard scoring requires agronomist review and status promotion to approved.

Catalog vs. Runtime Evidence

The catalog stores stable crop requirements and their references. FortyGuard and NASA provide location temperature evidence; SSURGO provides mapped texture, pH, drainage, and available-water-capacity evidence; farmer input supplies irrigation availability and can override mapped soil with a laboratory result. Forecasts, soil observations, scores, and rankings do not belong in a crop record.

Schema source: data/crop-catalog/catalog.schema.json. Catalog source: data/crop-catalog/catalog.json.